

Seis Magnitude

m_b - body wave magnitude - based on P-wave amplitude
 $m_b = \log_{10} (A/T) + Q(h, \Delta)$
 ← depth
 ← epicenter distance degrees
 ← empirical function based on earth responses.

M_s - surface wave magnitude - based on amplitude of surface waves
 $M_s = \log_{10} (A/T) + 1.66 \log_{10} \Delta + 3.3$ Rayleigh waves.

M_L - local magnitude
 Richter magnitude
 $M_L = \log_{10} A + 2.56 X - 1.67$
 ← displacement microns
 ← distance from source

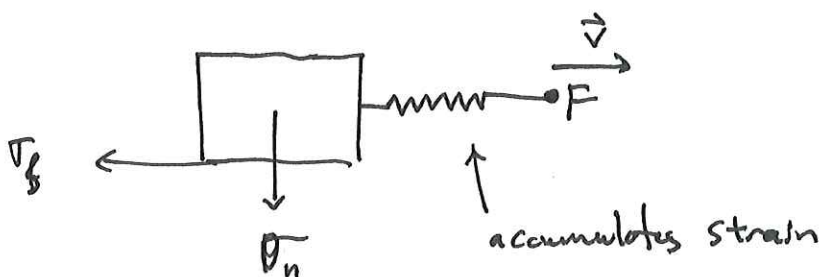
M_w - moment magnitude

$$M_w = \frac{2}{3} (\log_{10} M_0 - 9.1)$$

↑
earthquake moment

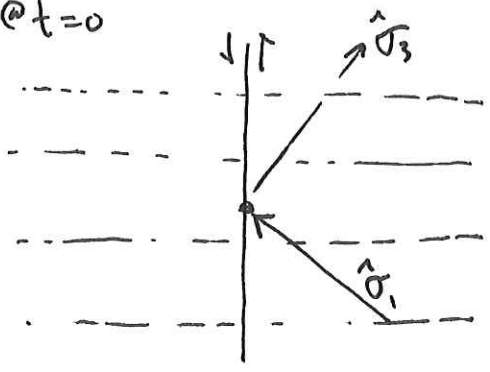
moment
 $M_0 = \mu \bar{D} A$
 ↑ ↑
 shear modulus displacement at fault during earthquake
 area at fault rupture

Earthquake analog

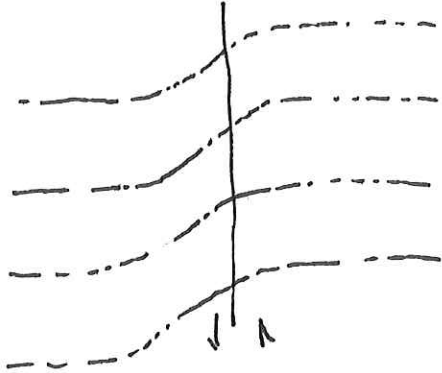


Earthquake Prediction / Forecasting

@ t=0

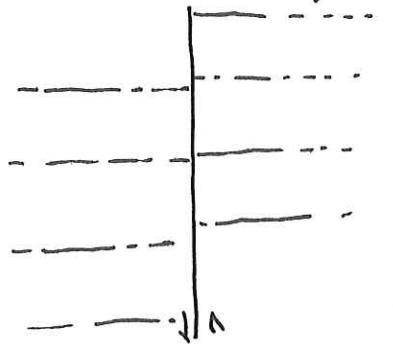


t > 0 before earthquake

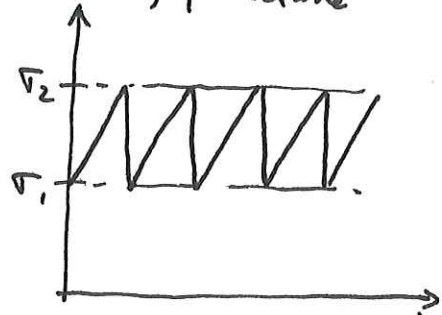


strain accumulates

t > 0 after earthquake



3 models for stress/strain completely predictable

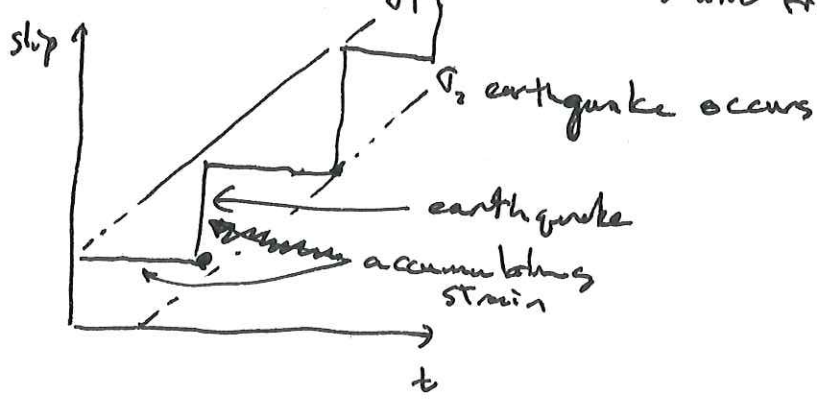


dynamic friction

$$\sigma_1 = \mu_d \sigma_n$$

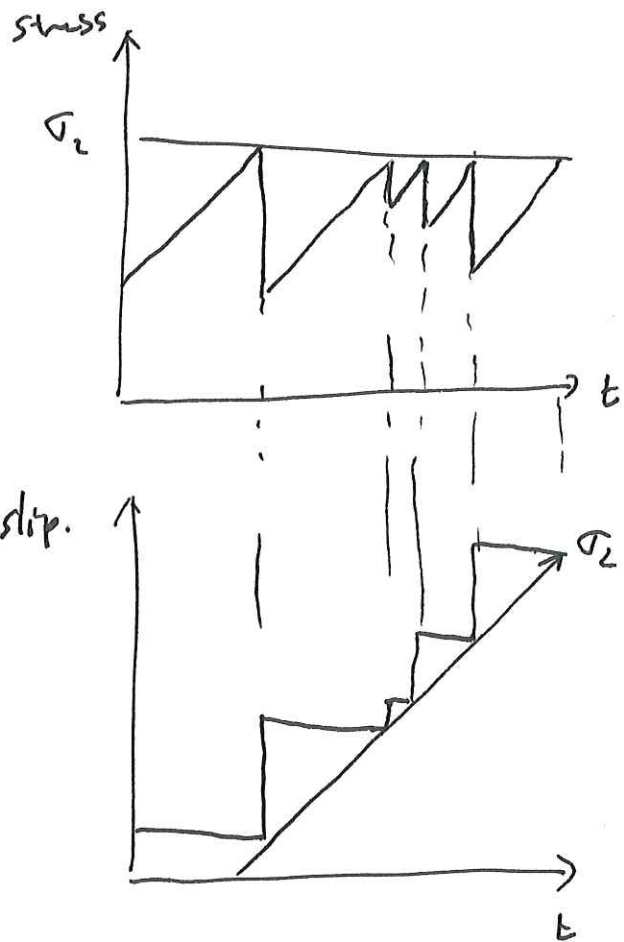
$$\sigma_2 = \mu_s \sigma_n$$

↑ ↑ normal stress
static friction

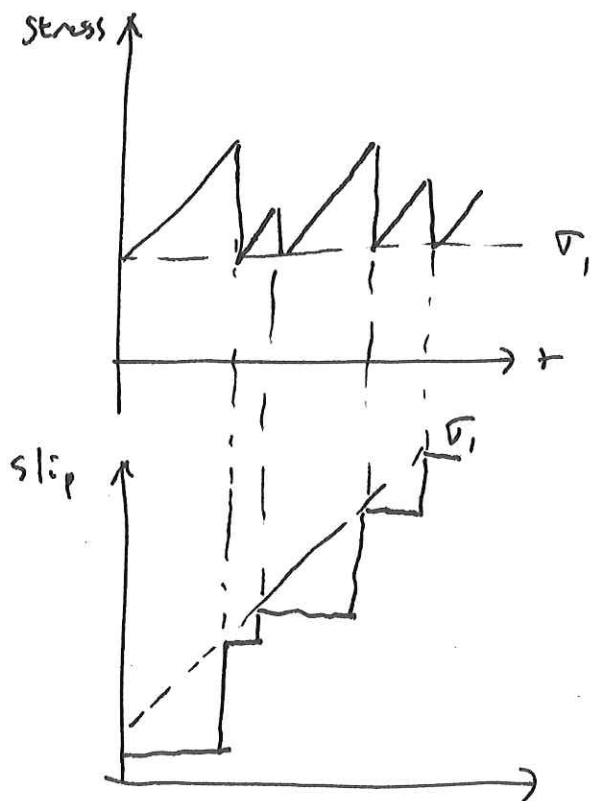


time predictable

time predictable



slip predictable



Cutenberg-Richter law

$$\log_{10} N = a - b M_w$$

↑
number of earthquakes

↑
property of region
typically 0.8 to 1.2

